



Instructions

- **Part A** (40 pts): Conceptual fill-in-blank, multiple choice, short answer.
- **Part B** (60 pts): Analysis & Application calculation, figure/scenario analysis.
- Total: **100 points**. Good luck!

PART A: CONCEPTUAL UNDERSTANDING

(40 points)

Fill in the Blank (5 questions, 20 points total, 4 points each)

A.1 RIGID REGISTRATION DoF

A 3D rigid registration has _____ degrees of freedom, consisting of _____ for rotation and _____ for translation.

A.2 SIMILARITY METRICS

Mutual Information (MI) as a similarity metric has the advantage that it can handle registration of images from _____ modalities (e.g. CT and MRI), whereas Sum of Squared Differences (SSD) is only suitable for _____ modality registration.

A.3 SELF-ATTENTION

In the self-attention mechanism, the input is mapped into three vectors: _____, _____, and _____. The attention weights are computed from the dot product of _____ and _____.

A.4 SEMI-SUPERVISED LEARNING

Semi-supervised learning simultaneously uses _____ data and _____ data for training. It is well-suited for medical imaging scenarios where _____ is expensive.

A.5 ACTIVE LEARNING

The core idea of active learning is to have the model select the most _____ samples for expert annotation, achieving the greatest performance improvement with the fewest possible labels.

Multiple Choice (3 questions, 9 points total, 3 points each)**A.6 RIGID REGISTRATION**

[] Rigid registration can handle which of the following geometric transformations?

- (A) Rotation + Translation + Scaling
- (B) Rotation and Translation only
- (C) Rotation + Translation + Shearing
- (D) Arbitrary non-linear deformations

A.7 MULTI-MODAL SIMILARITY METRIC

[] Which similarity metric is most suitable for multi-modal registration between CT and MRI?

- (A) Sum of Squared Differences (SSD)
- (B) Normalized Cross-Correlation (NCC)
- (C) Mutual Information (MI)
- (D) Mean Squared Error (MSE)

A.8 SELF-SUPERVISED LEARNING

[] In self-supervised learning, the model is trained by:

- (A) Using expert-annotated labels for training
- (B) Constructing pretext tasks from the data itself to learn representations
- (C) Jointly using a small amount of labeled data and a large amount of unlabeled data
- (D) Having the model automatically generate perfect labels for training

Short Answer (1 question, 11 points)**A.9 REGISTRATION TYPES COMPARISON**

Briefly compare rigid registration, affine registration, and deformable registration: what types of transformations does each allow? What scenarios are each best suited for? (No more than 150 words.)

PART B: ANALYSIS & APPLICATION

(60 points)

Calculation (1 question, 20 points)**B.1 2D RIGID TRANSFORMATION**

Given a 2D rigid transformation with rotation angle $\theta = 30^\circ$ and translation $(t_x, t_y) = (5, 3)$.

The transformation formula is:

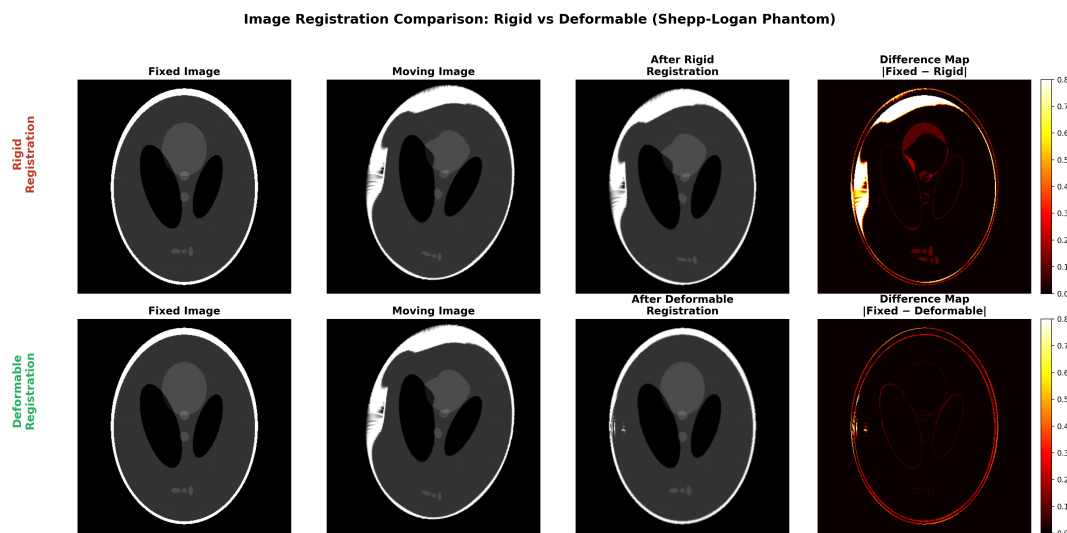
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

where $\cos 30^\circ \approx 0.866$ and $\sin 30^\circ = 0.5$.

- (6 pts) Write out the complete transformation matrix.
- (8 pts) Compute the new coordinates of point $P = (10, 0)$ after the transformation.
Show your calculation steps.
- (6 pts) If the rotation angle is changed to 0° , what operation does the transformation reduce to? Write out the simplified formula.

Figure / Structure Analysis (1 question, 20 points)**B.2 REGISTRATION COMPARISON**

The figure below shows the results of applying rigid registration and deformable registration to a Shepp–Logan phantom (a mathematical model simulating brain structure).



- (a) (6 pts) Compare the two registration results and identify the regions where rigid registration performs poorly.
- (b) (8 pts) Explain why deformable registration achieves better results in those regions.
- (c) (6 pts) Is deformable registration always better? Describe a scenario where deformable registration might produce unreasonable results.

Scenario Application (1 question, 20 points)

B.3 LABEL-EFFICIENT LEARNING FOR CHEST X-RAYS

Your research team has obtained a dataset of 10,000 chest X-ray images, but only 500 have physician annotations (with disease category labels). The remaining 9,500 are unlabeled. You need to build a disease classification model.

Here are three candidate learning paradigms:

- (A) Use only the 500 labeled images for fully supervised learning.
 - (B) Use semi-supervised learning, jointly leveraging labeled and unlabeled data.
 - (C) First perform self-supervised pre-training on all 10,000 images, then fine-tune with the 500 labeled images.
- (a) (12 pts) Which approach would you choose? Explain your reasoning.
 - (b) (8 pts) What is the fundamental difference between how approaches (B) and (C) utilize unlabeled data?